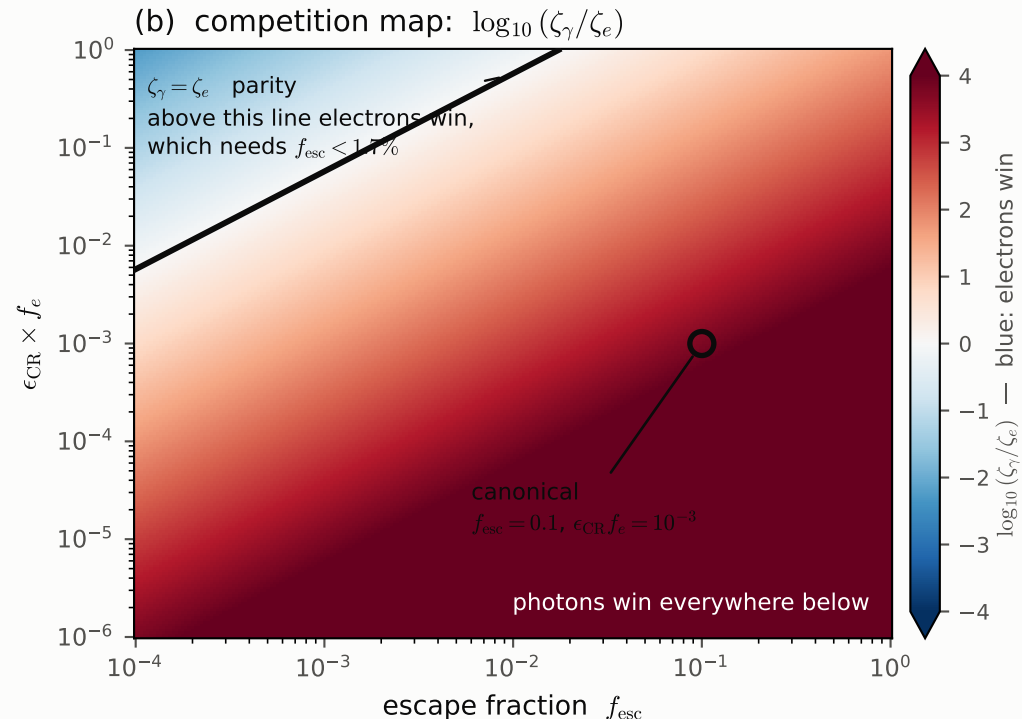
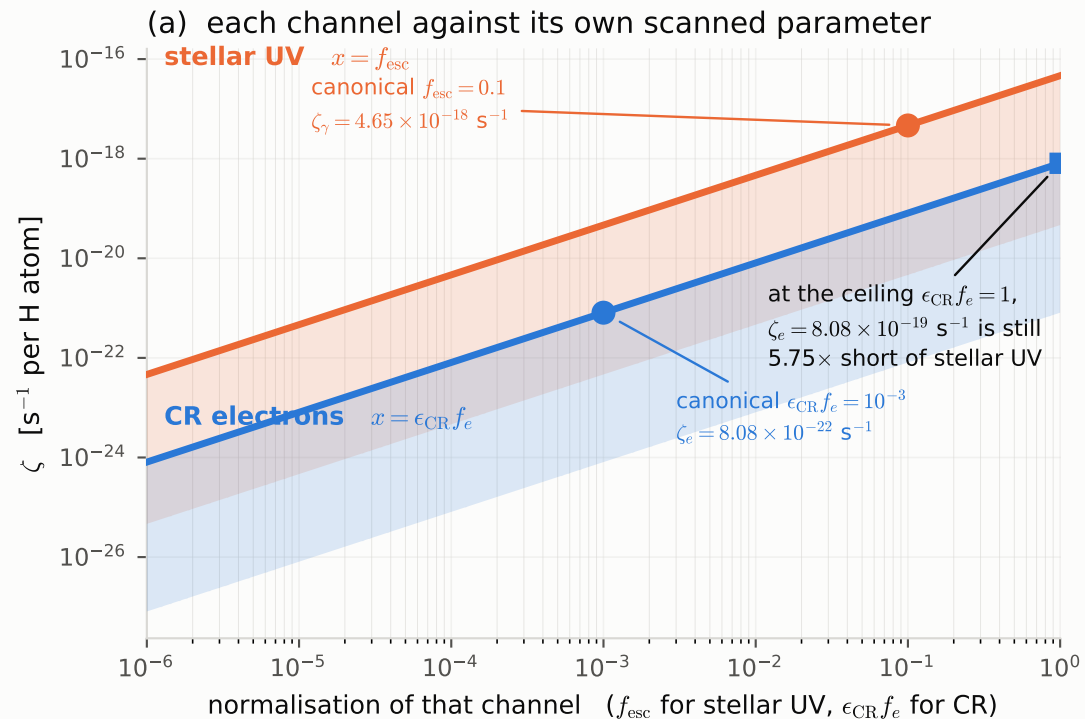


Parameter exploration: what would it take for CR electrons to compete?



IGM at $z=20$, $x_e = 10^{-4}$, pure H.

Electron yields: the loss+IC route — igm_losses.py: 7 loss mechanisms, RBED event counting, BED secondary cascade, Klein-Nishina IC, + IC-secondary photoionization.

ρ_{UV} : Donnan+24 JWST PRIMER. $\xi_{\text{ion}} = 10^{25.28}$: Llerena+25. $\rho_{\text{UV}} \rightarrow \rho_{\text{SFR}}$: Madau & Dickinson 14 (Salpeter).

SN/ $M_\odot$ derived from the IMF, not adopted.

Stellar SED: population of U37126 ($z=10.255$, Marques-Chaves+26, BPASS v2.2.1, $Z=0.003$, 6.8 Myr); LyC shape parametrised, α bracketed 1–3.

NOT ANCHORED: ρ_{UV} and ξ_{ion} are measured at $z \simeq 10$, not $z=20$. The shaded band spans $\rho_{\text{SFR}} \in [0.001, 1] \times$ its $z=10$ value — an ignorance interval, not an error bar. The RATIO is unaffected: ρ_{SFR} feeds both channels and cancels.

Left: both normalisations are dimensionless fractions, so both curves end at 10^0 — beyond it there is no more energy to give. Square marker: the CR channel at its absolute ceiling.